

Katherine Hutchinson

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EDUCATION

University of Virginia | Echols Scholar

May 2027

Bachelor of Computer Science | AI Focal Path | Minor in Data Science | 3.79 GPA

Coursework: AI, Machine Learning, Data Engineering, Computer Systems & Organization, Data Structures & Algorithms, Linear Algebra, Discrete Math

EXPERIENCE

Software Engineering Intern

May 2026 – August 2026

TELUS Digital

Charlottesville, VA

- Engineered PostgreSQL database migrations with Entity Framework Core to update schema and array data, improving data integrity for an ASP.NET application serving 200K+ users and millions of assessments annually.
- Developed a REST API for a new feature, implementing validation and database operations, writing unit and integration tests, and manually testing API calls in Swagger.
- Demoed tickets to clients, clearly explaining and demonstrating changes and alignments with requirements.

Software Engineering Intern

May 2025 – July 2025

Savannah River Nuclear Solutions

Aiken, SC

- Designed interactive virtual reality (VR) training modules for the Meta Quest 3 for motor-skill coordination proficiency to reduce VR training time for beginners.
- Developed a radiation dosimetry simulation using the inverse square law to model dose exposure, focusing on radiation safety and aligning with company goals.
- Automated Unity prefab generation through reusable C# editor scripts, eliminating repetitive manual setup and reducing scene creation time.

PROJECTS

New York Times Archives Data Analysis | Python | github.com/katiehutc/nytimes

- Processed 400K New York Times articles using Prefect orchestration, DuckDB, and BERTopic to generate unsupervised topic classifications based on semantic word meaning, reducing manual sorting.
- Built an interactive Streamlit analytics dashboard using DuckDB and Plotly to visualize long-term topic trends spanning 10 years; analyzed topic trends to uncover prioritization of news.

Course Reviews | Java, JavaFX, MySQL

- Developed a JavaFX/FXML application with JDBC integration to manage a relational database for UVA course reviews, implementing features such as user authentication, course management, and review submissions.
- Collaborated with team members using GitHub for version control, ensuring code consistency and minimizing integration errors.

EXTRACURRICULARS

Teacher's Assistant

Aug. 2026 – Present

University of Virginia

Charlottesville, VA

- TA for Discrete Math and Theory 2, assisting in homework grading, quiz grading, and holding office hours.

Hackathon Team Lead

Sep. 2024 – March 2025

Women in Computing Sciences

Charlottesville, VA

- Led workshop planning committee for club's annual hackathon, coordinating professional and recreational events, securing speakers from industry and academia, and managing scheduling logistics.

Analytics Desk Member

Feb. 2026 – present

The Cavalier Daily

Charlottesville, VA

- Analyzing and visualizing U.Va.'s oldest student newspaper's website engagement data using Google Analytics and SNWorks to support audience and content strategy decisions.

TECHNICAL SKILLS

Languages: Java, Python, C, C#, JavaScript, SQL, HTML/CSS

Frameworks and Tools: ASP.NET, EF Core, React, Tailwind CSS, Unity XR, JavaFX, Git, Swagger UI

Data and Cloud: PostgreSQL, DuckDB, Prefect, Pandas, Matplotlib, BERTopic, AWS S3